

Algorithm Used: Rule-Based Text Processing

This project does not utilize a machine learning algorithm such as Logistic Regression, Decision Trees, or Neural Networks. Instead, it applies a rule-based text processing approach using Python logic and library functions.

Why This Approach?

The goal of the project is to convert structured plain text into formatted PowerPoint slides. Since the transformation is deterministic — meaning that given the same input, the output will always be the same — a traditional ML algorithm is not necessary.

Key Characteristics of the Algorithm:

- **Pattern Matching:** Recognizes slide titles, bullet points, and paragraph breaks using newline patterns.
- **Text Wrapping:** Splits long lines after a fixed number of words (e.g., 10) for layout consistency.
- **User Interaction:** Uses UI elements (`ipywidgets`) to control input and output behavior.

Conclusion:

A rule-based method is the most appropriate algorithmic strategy for this problem, as it offers complete control over the structure, appearance, and behavior of the output without the overhead or unpredictability of training a model. It's efficient, interpretable, and perfectly suited for real-time user-driven applications like this one.